

Karan Jadhav

Senior Backend Engineer | Python, FastAPI, Django, PostgreSQL, AWS

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Summary

Senior Backend Engineer with 5+ years building production APIs, async services, database-backed systems, and AWS infrastructure. Strongest work includes FastAPI/Django REST APIs with p99 latency under 200 ms, distributed background pipelines processing 50K+ jobs/day, and PostgreSQL/PostGIS systems handling **2.3B+ records and 1.8 TB+ indexed data**.

Selected Impact

- Cut critical query/API latency from **minutes to single-digit milliseconds** through indexing, query planning, and caching.
- Designed and ran distributed pipelines processing **50K+ jobs/day** with retries, scheduling, and fault recovery.
- Held FastAPI/Django REST APIs to **p99 < 200 ms** under production load across enterprise customers.
- Built and owned PostgreSQL/PostGIS systems at **2.3B+ records / 1.8 TB+** indexed, plus a raster service serving **5.3 TB**.

Core Expertise

- Backend APIs: Python, FastAPI, Django, REST design, validation, pagination, async services.
- Database-Backed Systems: PostgreSQL, PostGIS, Redis, indexing, query plans, schema design.
- Async Processing: RabbitMQ, Dask, worker orchestration, retries, scheduling, failure handling.
- Performance: p99 latency, caching, query tuning, throughput improvement, database load reduction.
- Cloud Backend: AWS EC2, S3, RDS, SQS, Lambda, CloudWatch, Docker, CI/CD.
- Production Reliability: structured logging, monitoring, deployment, debugging, incident support.

Experience

Software Engineer (Backend & Data Infrastructure) - Intensel Ltd | 2021 - Present

- Owned backend systems end to end — API design, distributed processing, database performance, and AWS infrastructure — for a production climate-risk platform used by enterprise customers across 10+ countries.
- Reduced critical API/query latency from minutes to single-digit milliseconds via indexing, query-plan tuning, and cache-aside strategies, materially cutting database load.
- Built high-performance FastAPI and Django REST APIs sustaining p99 latency under 200 ms under production load.
- Designed and operated distributed background job pipelines processing 50K+ jobs/day on RabbitMQ and Dask, with retries, scheduling, worker orchestration, and failure recovery.
- Designed backend services over PostgreSQL/PostGIS datasets with 2.3B+ geospatial records and 1.8 TB of indexed data.
- Managed AWS infrastructure (EC2, S3, RDS, SQS, Lambda, CloudWatch) and shipped CI/CD plus observability that cut deployment time by 60%.
- Added structured logging and monitoring that improved production debugging and incident response.

Projects

Global Building Footprints

- Built backend ingestion and query infrastructure for 2.3B+ building records.
- Enabled low-latency analytics workflows by optimizing API/database access paths.

Map Tile Service

- Designed scalable backend services for geospatial raster delivery serving 5.3 TB of data.
- Improved throughput and latency through caching and performance tuning.

Open Source

RediServe | Rust, Axum, Tokio

- Built an async HTTP API for Redis with connection pooling.
- Focused on low-latency request handling and efficient backend resource usage.

Technical Skills

Languages: Python, Rust, Go, JavaScript, TypeScript, SQL

Backend: FastAPI, Django, REST APIs, async workers, API design

Databases: PostgreSQL, PostGIS, Redis, DuckDB

Cloud: AWS EC2, S3, RDS, SQS, Lambda, CloudWatch

Infrastructure: Docker, Linux, CI/CD, observability

Systems: distributed systems, message queues, caching, scalability

Education

Bachelor of Computer Science - University of Mumbai | 2020